

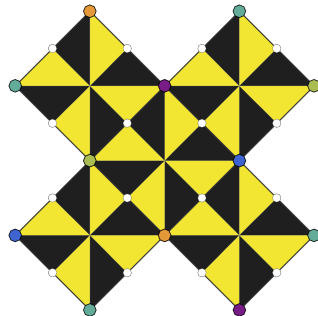
Minimal Chiral Polyhedra

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QUESTIONS

How can we leverage the characteristics of families of chiral structures to solve certain classification problems?

Can we reduce the problem of building complicated chiral structures to fashioning simpler structures on top of basal ones?

PART 1

Background

1.0 ABSTRACT POLYHEDRA

1.1 Motivation & definitions

1.2 Automorphisms & coverings

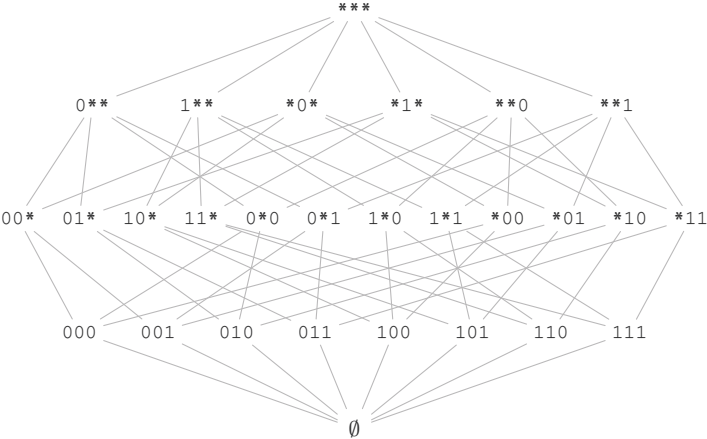
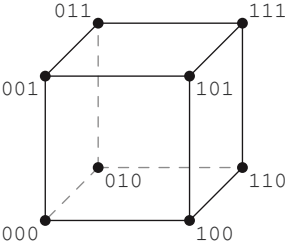
2.0 HIGHLY SYMMETRIC POLYHEDRA

2.1 Regular polyhedra

2.2 Chiral polyhedra

MOTIVATION

Abstract polyhedra

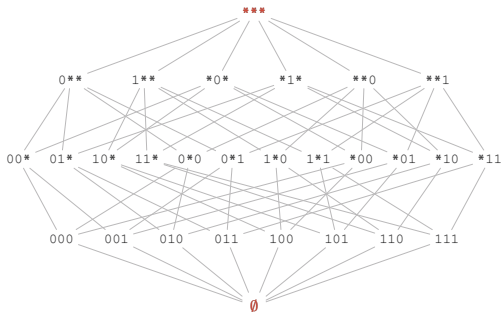


DEFINITION

Abstract polyhedra

An abstract polyhedron is a poset $(\mathcal{P}, \preceq)$ satisfying:

(P1) There's a min, max face m, M .

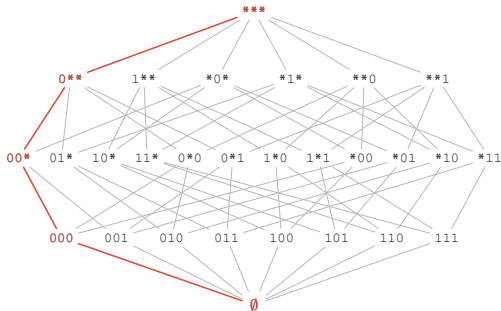


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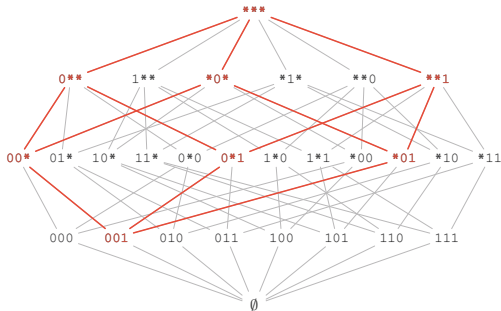
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is connected.



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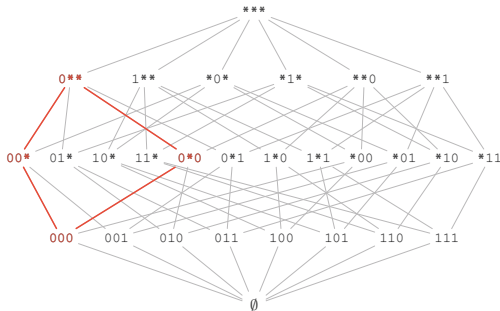
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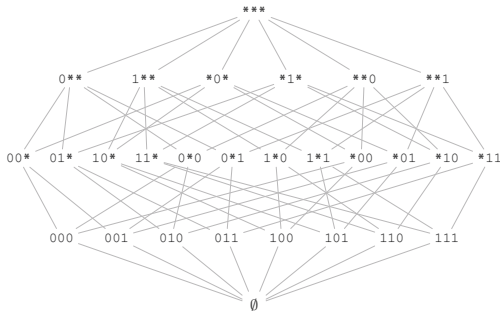
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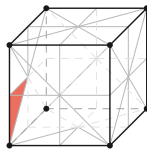
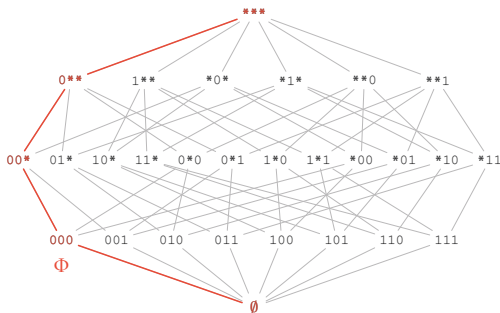
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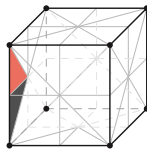
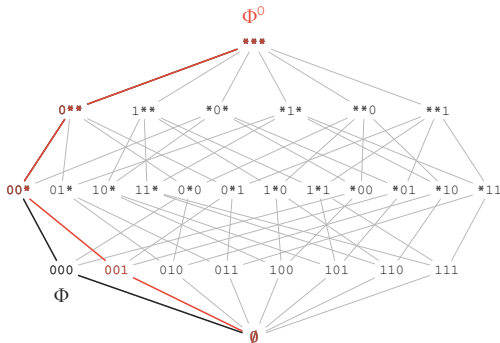
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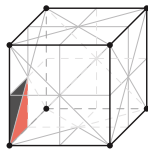
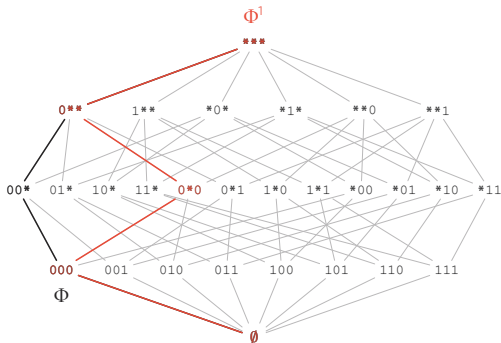
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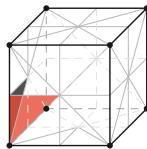
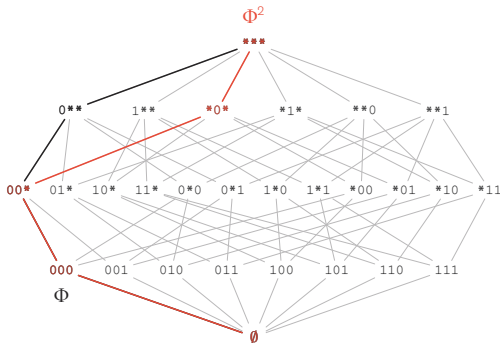
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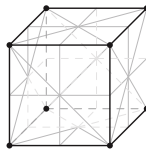
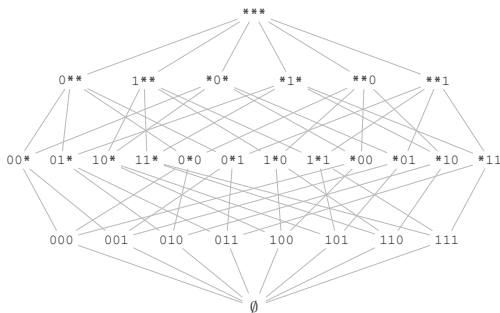
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is **flag-connected**.

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Abstract polyhedra

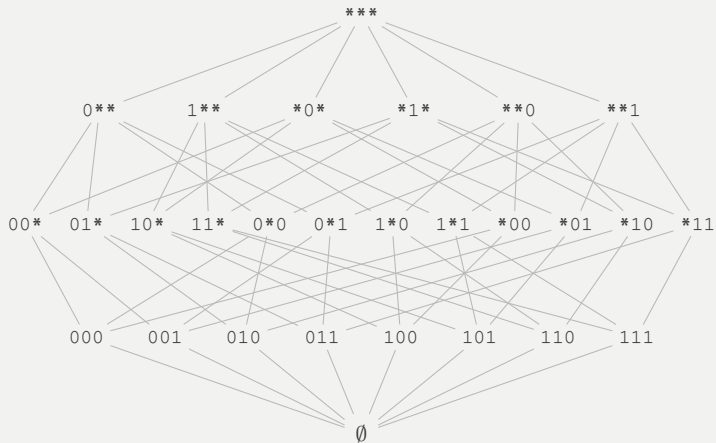
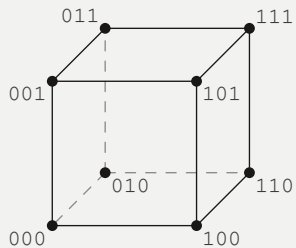
Conventions

- The 0-, 1-, and 2-faces are the vertices, edges, and facets.
- Each face F can be identified with the section F/m .
- The co-face at F is the section M/F . When F is a vertex, this is the vertex figure.

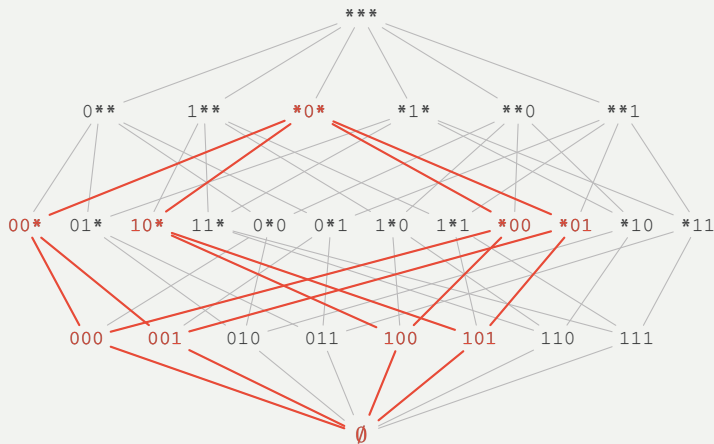
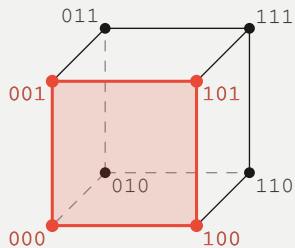
Definition (Schläfli symbol)

If every facet of $\mathcal{P}$ is a t_1 -gon and every vertex figure is a t_2 -gon, then $\mathcal{P}$ is equivelar and is assigned a type given by the Schläfli symbol $\{t_1, t_2\}$.

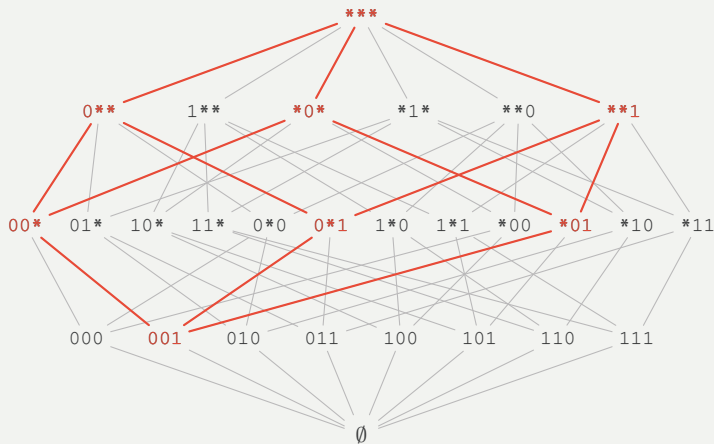
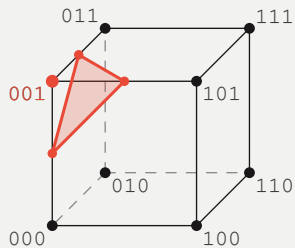
Example (The cube is of type {4, 3})



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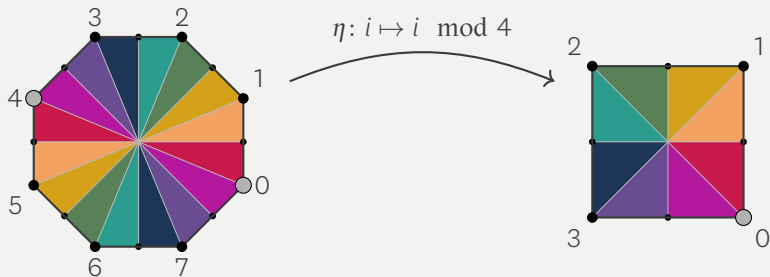
Homomorphism

Let $\mathcal{P}, \mathcal{Q}$ be abstract polyhedra.

- A homomorphism is an incidence-preserving map $\mathcal{P} \rightarrow \mathcal{Q}$.
- An automorphism of $\mathcal{P}$ is a bijective homomorphism $\mathcal{P} \rightarrow \mathcal{P}$.
- The collection $\Gamma(\mathcal{P})$ of automorphisms of $\mathcal{P}$ forms the group of $\mathcal{P}$.
- A duality of $\mathcal{P}$ is an incidence-reversing bijection $\mathcal{P} \rightarrow \mathcal{P}$.
- A covering is a rank- and adjacency-preserving surjection $\mathcal{P} \rightarrow \mathcal{Q}$.

Notation: $\mathcal{P} \searrow \mathcal{Q}$.

Example (The octagon covers the square)



DEFINITION

Regular polyhedra

If every flag of $\mathcal{P}$ is equivalent under $\Gamma(\mathcal{P})$, then $\mathcal{P}$ is regular.

Let $\mathcal{P}$ be regular and Φ a flag of $\mathcal{P}$.

■ There exist involutions $\rho_0, \rho_1, \rho_2 \in \Gamma(\mathcal{P})$ characterized by

$$\Phi \rho_i \stackrel{\text{def}}{=} \Phi^i \quad (i \in \{0, 1, 2\}).$$

Notation: $\mathcal{P}_{(\Phi)}$ when Φ is the fixed, base flag.

■ $\Gamma(\mathcal{P}_{(\Phi)}) = \langle \rho_0, \rho_1, \rho_2 \rangle$, and has at least the relations

$$\rho_0^2 = \rho_1^2 = \rho_2^2 = (\rho_0 \rho_1)^s = (\rho_0 \rho_2)^2 = (\rho_1 \rho_2)^t = \varepsilon.$$

■ So $\mathcal{P}$ is of type $\{s, t\}$, and $\Gamma(\mathcal{P}) \cong [s, t]/N$ for some $N \trianglelefteq [s, t]$.

Regular polyhedra

Let $\mathcal{U}, \mathcal{P}$ be abstract polyhedra with $\mathcal{U}$ regular.

If $\mathcal{U} \searrow \mathcal{P}$, then $\mathcal{P}$ is regular iff $\Gamma(\mathcal{P}) \cong \Gamma(\mathcal{U})/N$ for $N \trianglelefteq \Gamma(\mathcal{U})$.

Convention (McMullen, Schulte [1, Proposition 2D8])

Write $\mathcal{P} = \mathcal{U}/N$ as the set of orbits of N in $\mathcal{U}$. If $\mathcal{P}$ is a polyhedron, then

$$\mathcal{P} \text{ regular} \iff \Gamma(\mathcal{U})/N \cong \Gamma(\mathcal{U}/N) = \Gamma(\mathcal{P}) \quad (N \trianglelefteq \Gamma(\mathcal{U})).$$

The regular $\mathcal{P}$ of type $\{s, t\}$ are characterized by the string C-groups of type $\{s, t\}$.

DEFINITION

Chiral polyhedra

If $\Gamma(\mathcal{P})$ induces exactly 2 orbits on the flags with adjacent flags in opposite orbits, $\mathcal{P}$ is chiral.

Let $\mathcal{P}$ be chiral and Φ a flag of $\mathcal{P}$.

■ There exist rotations $\sigma_1, \sigma_2 \in \Gamma(\mathcal{P})$ characterized by

$$\Phi\sigma_i = \Phi^{i-1, i} \stackrel{\text{def}}{=} (\Phi^{i-1})^i \quad (i \in \{1, 2\}).$$

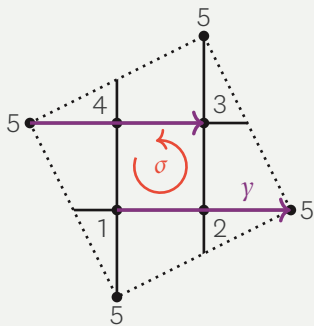
Note: In the regular case, $\sigma_i = \rho_{i-1}\rho_i$.

■ $\Gamma(\mathcal{P}_{(\Phi)}) = \langle \sigma_1, \sigma_2 \rangle$, and has at least the relations

$$\sigma_1^s = \sigma_2^t = (\sigma_1\sigma_2)^2 = \varepsilon.$$

■ So $\mathcal{P}$ is of type $\{s, t\}$, and $\Gamma(\mathcal{P}) \cong [s, t]^+ / N$ for some $N \trianglelefteq [s, t]^+$.

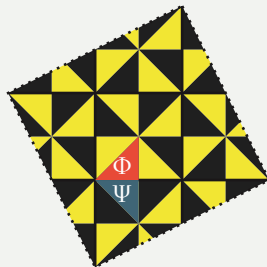
Example (The chiral map $\{4, 4\}_{(1,2)}$)



(a)

$$\sigma = (1\ 2\ 3\ 4)$$

$$\gamma = (1\ 2\ 5\ 4\ 3)$$



(b)

$$\Phi\rho = \Psi, \quad \rho^2 = \varepsilon$$

$$\rho: 3 \mapsto 4 \mapsto 5$$

Chiral polyhedra

Let $\mathcal{U}, \mathcal{P}$ be abstract polyhedra with $\mathcal{U}$ regular.

If $\mathcal{U} \searrow \mathcal{P}$, then $\mathcal{P}$ is chiral iff $\Gamma(\mathcal{P}) \cong \Gamma(\mathcal{U})/N$ for some $N \trianglelefteq \Gamma^+(\mathcal{U})$ with $N \not\trianglelefteq \Gamma(\mathcal{U})$

The chiral $\mathcal{P}$ of type $\{s, t\}$ are characterized by the string C^+ -groups of type $\{s, t\}$.

Convention

Suppose $\mathcal{P}$ is chiral. Then $\mathcal{P}$ has two enantiomorphs, $\mathcal{P}$ and $\mathcal{P}^*$.

- WLOG, $\mathcal{P} = \mathcal{P}_{(\Phi)}$ and $\mathcal{P}^* = \mathcal{P}_{(\Phi^0)}$ for some base flag Φ of $\mathcal{P}$.
- If $\Gamma(\mathcal{P}_{(\Phi)}) = \langle \sigma_1, \sigma_2 \rangle$, then $\Gamma(\mathcal{P}_{(\Phi^0)}) \cong \langle \sigma_1^{-1}, \sigma_1^2 \sigma_2 \rangle$.

Theorem (Schulte, Weiss [2, Theorem 1])

Let $\Gamma = \langle \sigma_1, \sigma_2 \rangle$ be a group with $o(\sigma_1) = s$, $o(\sigma_2) = t$ for some $s, t \geq 3$, and $o(\sigma_1\sigma_2) = 2$. Also suppose $\langle \sigma_1 \rangle \cap \langle \sigma_2 \rangle = \{\varepsilon\}$. Then Γ is the group of a chiral polyhedron $\mathcal{P}(\Gamma)$ of type $\{s, t\}$ iff there is no involution $\rho \in \text{Aut } \Gamma$ mapping

$$\rho(\sigma_i) = \begin{cases} \sigma_1^{-1}, & \text{if } i = 1; \\ \sigma_1^2 \sigma_2, & \text{if } i = 2. \end{cases}$$

Upshot

We can construct chiral polyhedra from a pair of group elements $\sigma_1, \sigma_2 \in \Gamma$ satisfying:

Generation $\Gamma = \langle \sigma_1, \sigma_2 \rangle$.

Intersection condition $\langle \sigma_1 \rangle \cap \langle \sigma_2 \rangle = \{\varepsilon\}$.

Chirality There is no involution $\rho \in \text{Aut } \Gamma$ mapping $\rho(\sigma_1) = \sigma_1^{-1}$ and $\rho(\sigma_2) = \sigma_1^2 \sigma_2$.

PART 2

Classification

3.0 BASAL POLYHEDRA

3.1 Motivation & definition

4.0 BASAL POLYHEDRA IN $\mathcal{C}(\mathbb{Z}^+, 1)$

4.1 Application: chiral polyhedra of type $\{2^s, 2^t\}$

4.2 Application: chiral polyhedra of type $\{2^s p, 2^t p\}$

5.0 CHIRAL POLYHEDRA FROM $\text{AGL}(1, q)$

Question

Are there group structures which are characteristic to certain families of chiral polyhedra?

Approach

Let $\mathcal{C}$ be a collection of chiral polyhedra. If there is a covering sequence

$$\cdots \searrow \mathcal{P}_2 \searrow \mathcal{P}_1 \searrow \mathcal{P}_0 \quad (\mathcal{P}_j \in \mathcal{C} \text{ for all } j \in \mathbb{N})$$

where no $\mathcal{P} \in \mathcal{C}$ is properly covered by $\mathcal{P}_0$, then $\Gamma(\mathcal{P}_j) = \Omega_{i,j} \cdot \Gamma(\mathcal{P}_i)$ for all $i \leq j$. Specifically,

$$\Gamma(\mathcal{P}_j) = \Omega_j \cdot \Gamma(\mathcal{P}_0) \quad (\Omega_j \stackrel{\text{def}}{=} \Omega_{j,0} \text{ for all } j \in \mathbb{N})$$

Definition (Basal polyhedron)

A chiral polyhedron $\mathcal{P}$ is basal (with respect to some $\mathcal{C}$) if $\mathcal{P} \searrow \mathcal{Q} (\in \mathcal{C})$ implies $\mathcal{P} = \mathcal{Q}$.

Tight chiral polyhedra

A chiral polyhedron $\mathcal{P}$ of type $\{t_1, t_2\}$ is tight if $|\Gamma(\mathcal{P})| = t_1 t_2$.

■ Denote by $\mathcal{T}$ the collection of tight chiral polyhedra.

Proposition (Cunningham [3, Theorems 3.6–8])

The first three families of basal $\mathcal{P}$ in $\mathcal{T}$ are of type...

- (i) $\{m^\alpha, 2m\}$ for m an odd prime and $\alpha \geq 2$.
- (ii) $\{2^{\beta-1}, 2^\beta\}$ for $\beta \geq 5$.
- (iii) $\{8, 2^\beta\}$ for $\beta \geq 5$.

Theorem (Cunningham [3, Theorem 5.2])

Up to duality, there is a tight chiral polyhedron of type $\{p, q\}$ iff one of the following is true:

- (i) p odd, q an even divisor of $2p$, and there is an odd prime m dividing q with m^2 dividing p .
- (ii) p, q both even, and there is an odd prime m dividing p with m^2 dividing q .
- (iii) 8 divides p , and 32 divides q .

Problem

Classify the groups of chiral polyhedra of order 2^n and $2^n p$.

Order type	Small groups	Chiral polyhedra
Total	49 910 531 351	4607
2^n	49 497 920 262	111
$2^n p$	412 055 081	855
$2^n p^m$ ($m > 1$)	249 437	1092

Basal polyhedra in $\mathcal{C}(\mathbb{Z}^+, 1)$

$\mathcal{C}(\mathbb{Z}^+, 1) \stackrel{\text{def}}{=} \{\text{chiral } \mathcal{P} \mid |\Gamma(\mathcal{P})| = 2^n p \text{ for some } n \in \mathbb{Z}^+ \text{ and odd prime } p\}.$

■ $\mathcal{P} \searrow \mathcal{Q} \text{ in } \mathcal{C}(\mathbb{Z}^+, 1) \implies \mathcal{Q} = \mathcal{P}/N \text{ for some 2-group } N \trianglelefteq \Gamma(\mathcal{P}).$

Corollary

A polyhedron $\mathcal{P}$ is basal in $\mathcal{C}(\mathbb{Z}^+, 1)$ if $\Gamma(\mathcal{P})$ has no non-trivial normal 2-subgroups.

Lemma

Let Γ be a group of order $2^n p$, and suppose $p \equiv 1 \pmod{2^n}$. Then

- (i) Γ has a normal Sylow p -subgroup, and $\Gamma \cong C_p \rtimes \Omega$ for any Sylow 2-subgroup Ω .*
- (ii) If Γ contains no non-trivial normal 2-subgroups, then every 2-subgroup of Γ is cyclic.*

Basal polyhedra in $\mathcal{C}(\mathbb{Z}^+, 1)$ **Proposition**

The following are equivalent for a basal polyhedron $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ of order $2^n p$:

- (i) $p \equiv 1 \pmod{2^n}$.
- (ii) $\mathcal{P}$ is of type $\{-, 2^n\}$ or $\{2^n, -\}$.
- (iii) $\mathcal{P}$ is of type $\{2^n, 2^n\}$.

Theorem

For each $n \geq 2$ and odd prime p satisfying $p \equiv 1 \pmod{2^n}$, there are exactly 2^{n-2} basal chiral polyhedra $\mathcal{P}$ of order $2^n p$ —those of type $\{2^n, 2^n\}$ and group $\Gamma(\mathcal{P}) \cong C_p \rtimes C_{2^n}$.

Polyhedra in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^s, 2^t\}$

So, $\mathcal{Q}$ is basal in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^n, 2^n\} \iff \Gamma(\mathcal{Q}) \cong C_p \rtimes C_{2^n}$ for some $p \equiv 1 \pmod{2^n}$.

Theorem

Every chiral polyhedron $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^s, 2^t\}$ covers exactly one basal polyhedron, namely $\mathcal{Q} = \mathcal{P}/O_2(\Gamma(\mathcal{P}))$ where $O_2(\Gamma(\mathcal{P}))$ is the 2-core of $\Gamma(\mathcal{P})$.

Definition (The mixing operation)

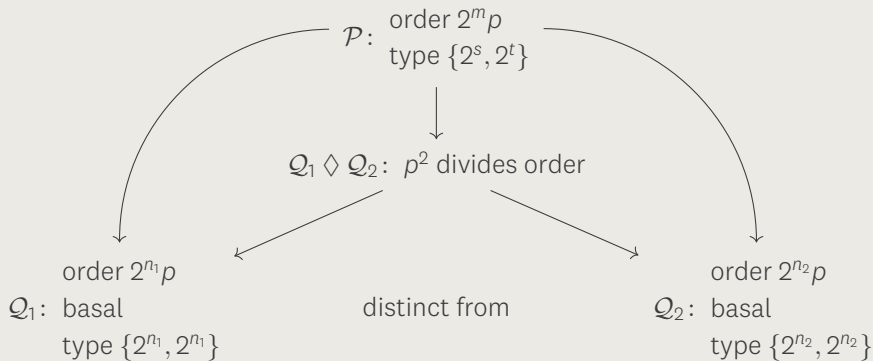
The mix of chiral polyhedra $\mathcal{P}, \mathcal{Q}$ is the smallest structure $\mathcal{P} \diamond \mathcal{Q}$ covering both $\mathcal{P}, \mathcal{Q}$.

Lemma

For each $i \in \{1, 2\}$, let $\mathcal{Q}_i$ be a basal of order $2^{n_i} p_i$ where $p_i \equiv 1 \pmod{2^{n_i}}$. If $\mathcal{Q}_1 \neq \mathcal{Q}_2$, then $p_1 p_2$ divides $|\Gamma(\mathcal{Q}_1 \diamond \mathcal{Q}_2)|$.

Polyhedra in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^s, 2^t\}$

Proof of uniqueness of basal $\mathcal{Q}$ covered by $\mathcal{P} \in \mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^s, 2^t\}$.



Constructing $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^s, 2^t\}$

...is really easy!

$$\{\varepsilon\} \longrightarrow \text{some 2-group } \Omega \longrightarrow \Gamma = \langle \tilde{\sigma}_1, \tilde{\sigma}_2 \rangle \longrightarrow \underbrace{C_p \rtimes C_{2^n} = \langle \sigma_1, \sigma_2 \rangle}_{\text{basal group}} \longrightarrow \{\varepsilon\}$$

Example (Trivial extensions by cyclic 2-groups)

Let $C_p \rtimes C_{2^n} = \langle \sigma_1, \sigma_2 \rangle$ be the group of some basal $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^n, 2^n\}$.
For each $m \geq 0$, take $\Omega = C_{2^m} = \langle x \rangle$. Then

$$\Omega \times (C_p \rtimes C_{2^n}) \cong \langle (\varepsilon, \sigma_1), (x, \sigma_2) \rangle$$

is the group of a chiral polyhedron in $\mathcal{C}(\mathbb{Z}^+, 1)$ of type $\{2^n, 2^{\max(m, n)}\}$.

Returning to $\mathcal{C}(\mathbb{Z}^+, 1)$...**Lemma**

Let Γ be a group of order $2^n p$. If $2^n \equiv 1 \pmod{p}$, then Γ has a unique Sylow 2-subgroup Ω . Furthermore,

- (i) If $\Gamma \cong \Omega \rtimes C_p$ is realized by some non-trivial action, then Ω is elementary abelian.
- (ii) If $n = \text{ord}_p(2)$, then $\mathbb{F}_2^n \rtimes C_p$ has no non-trivial normal 2-subgroups.

Fact: If such a non-trivial semidirect product $\Omega \rtimes C_p$ is the group of an equivelar polyhedron, then its type is $\{p, p\}$.

Proposition

Let $\mathcal{P}$ be a polyhedron of order $2^{\text{ord}_p(2)} p$ and type $\{p, p\}$. Then $\mathcal{P}$ is chiral iff $\text{ord}_p(2)$ is odd.

Returning to $\mathcal{C}(\mathbb{Z}^+, 1)$...**Theorem**

Fix p an odd prime with $\text{ord}_p(2)$ odd, and let $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ be of type $\{2^s p, 2^t p\}$.

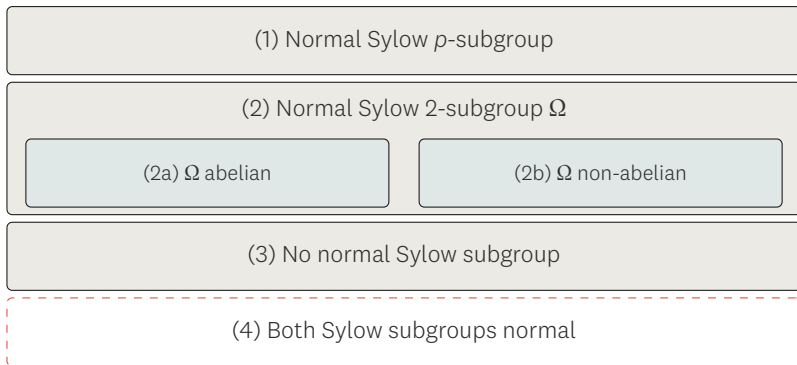
- (i) $\Gamma(\mathcal{P})$ has a unique Sylow 2-subgroup Ω .
- (ii) For some $j \in \{1, \dots, (p-1)/\text{ord}_p(2)\}$, $\mathcal{P}$ covers a unique j -fold mix of distinct basal polyhedra, each of type $\{p, p\}$

Specifically, for some $r \in \{0, 1, 2\}$,

$$\Gamma(\mathcal{P}) / \text{Frat } \Omega \cong C_2^r \times \left(\prod_{i=1}^j C_2^{\text{ord}_p(2)} \rtimes C_p \right) \cong C_2^r \times \Gamma(\underbrace{\{p, p\} \diamond \cdots \diamond \{p, p\}}_{j \text{ many, distinct, basal}}).$$

Returning to $\mathcal{C}(\mathbb{Z}^+, 1)$...

A basal polyhedron $\mathcal{P}$ in $\mathcal{C}(\mathbb{Z}^+, 1)$ of order $2^n p$ falls into one of the following cases:



Returning to $\mathcal{C}(\mathbb{Z}^+, 1)$...

Type	Group	Number	Notes	Basal?
$\{2^n, 2^n\}$	$C_p \rtimes C_{2^n}$	2^{n-2}	$p \equiv 1 \pmod{2^n}$	✓
$\{2^s, 2^t\}$	$C_p \rtimes \langle C_{2^n}, \Omega \rangle$	∞	Ω a 2-group	✗
$\{p, p\}$	$C_2^{\text{ord}_p(2)} \rtimes C_p$	$(p-1)/\text{ord}_p(2)$	p s.t. $\text{ord}_p(2)$ odd	✓
$\{2^{\varepsilon_1}p, 2^{\varepsilon_2}p\}$	$C_2^{\varepsilon_1+\varepsilon_2} \times (\prod_{i=1}^j C_2^{\text{ord}_p(2)} \rtimes C_p)$	$\binom{(p-1)/\text{ord}_p(2)}{j}$	$\varepsilon_i \in \{0, 1\}$, $j \in [(p-1)/\text{ord}_p(2)]$	✗
$\{2^s p, 2^t p\}$	$\Omega \cdot [C_2^r \times (\prod_{i=1}^j C_2^{\text{ord}_p(2)} \rtimes C_p)]$	∞	$r \in \{0, 1, 2\}$ $j \in [(p-1)/\text{ord}_p(2)]$	✗
???	$\Omega \rtimes C_p$	???	Ω non-abelian $p = 3?$	✓
$\{2^s p, 2^t\}$	No unique Sylow	???	Covers $\{p, 2\}$ $p = 3?$	✓

Chiral polyhedra from $\text{AGL}(1, \mathbb{F}_q)$

The affine group over the field $\mathbb{F}_q$ of $q = p^n$ elements is the group $\text{AGL}(1, \mathbb{F}_q)$ of maps

$$\alpha(a, b): \mathbb{F}_q \longrightarrow \mathbb{F}_q, \quad \alpha(a, b)(x) = ax + b$$

for $a \in \mathbb{F}_q^\times$ and $b \in \mathbb{F}_q$.

The group law in $\text{AGL}(1, \mathbb{F}_q)$ is

$$\alpha(a, b) \alpha(c, d) = \alpha(ac, ad + b).$$

■ Involutions in $\text{AGL}(1, \mathbb{F}_q)$ are characterized by $\alpha(-1, b)$.

Chiral polyhedra from $\text{AGL}(1, \mathbb{F}_q)$

Approach

For choice $\sigma_2 = a(a, b)$ and $\tau = a(-1, c)$, can recover σ_1 from $\sigma_1\sigma_2 = \tau \implies \sigma_1 \stackrel{\text{def}}{=} \tau\sigma_2^{-1}$:

$$\sigma_1 = a(-a^{-1}, a^{-1}b + c).$$

Then check generation, intersection condition, and chirality separately, modifying the choices for a, b, c as we go.

Verifying the intersection condition

Want to show $\langle \sigma_1 \rangle \cap \langle \sigma_2 \rangle = \{\varepsilon\}$. Suppose $\sigma_1^i = \sigma_2^j$ for some i, j . Direct computation gives

$$\sigma_1^i = a\left((-a^{-1})^i, \frac{(-a^{-1})^i - 1}{(-a^{-1}) - 1}(a^{-1}b + c)\right) = a\left(a^j, \frac{a^j - 1}{a - 1}b\right) = \sigma_2^j.$$

Some algebra rules out $c = 2b/(1 - a)$.

Chiral polyhedra from $\text{AGL}(1, \mathbb{F}_q)$

Theorem (Angelone, Schulte [4, Theorem 4.1])

Let $q \neq 3$ be an odd prime power, $a \in \mathbb{F}_q^\times \setminus \{\pm 1\}$, $b \in \mathbb{F}_q$, and $c \in \mathbb{F}_q \setminus \{2b/(1-a)\}$. Set

$$\sigma_1 = \alpha(-a^{-1}, a^{-1}b + c) \quad \text{and} \quad \sigma_2 = \alpha(a, b).$$

(i) $\Gamma \stackrel{\text{def}}{=} \langle \sigma_1, \sigma_2 \rangle \leq \text{AGL}(1, \mathbb{F}_q)$ is the group of a chiral polyhedron of type $\{o(-a^{-1}), o(a)\}$.

(ii) Suppose $\mathbb{F}_q^\times = \langle y \rangle$ where $a = y^m$ for some $1 \leq m < q-1$. Then

$$\Gamma = \text{AGL}(1, \mathbb{F}_q) \iff \begin{array}{l} \gcd(q-1, m) = 1, \quad \text{or} \\ \gcd(q-1, m) = 2 \text{ and } q \equiv 3 \pmod{4}. \end{array}$$

Chiral polyhedra from $\text{AGL}(1, \mathbb{F}_q)$ **Example** ($p \equiv 1 \pmod{2^n}$, $n \geq 2$ **maximal**)

Fix $p \equiv 1 \pmod{2^n}$ for some maximal $n \geq 2$.

- $\mathbb{F}_p^\times$ is cyclic of order $p - 1 = 2^n k$ for some odd $k \in \mathbb{Z}^+$.
- There's a unique subgroup $\langle a \rangle \cong C_{2^n}$ of $\mathbb{F}_p^\times$. (WLOG, $a \neq \pm 1$)
- Then $o(a) = o(-a^{-1}) = 2^n$.
- For any $b, c \in \mathbb{F}_p$ with $c \neq 2b/(1 - a)$,

$$\Gamma \stackrel{\text{def}}{=} \langle a(-a^{-1}, a^{-1}b + c), a(a, b) \rangle \leq \text{AGL}(1, \mathbb{F}_p)$$

is the group of a chiral polyhedron of order $2^n p$ and type $\{2^n, 2^n\}$, and $\Gamma \cong C_p \rtimes C_{2^n}$.

- $\Gamma = \text{AGL}(1, \mathbb{F}_p)$ iff p is Fermat; otherwise, $[\text{AGL}(1, \mathbb{F}_p) : \Gamma] = k$.

PART 3

Closing thoughts

6.0 FUTURE DIRECTIONS

6.1 Short term goals

6.2 Medium term goals

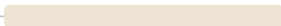
6.3 Long term goals



Short term



Medium term



Long term

Relating to $[3, 5]$...

Conjecture (Chiral polyhedra from $\text{AGL}(1, \mathbb{Z}_q)$)

Fix $q = p^{m-1}$ for some $m \geq 3$, and let $a = 1 + p^{m-2}$, $b \in \text{U}(q)$, and $c \in \mathbb{Z}_q$. Define $\sigma_1 = \alpha(a, b)$, $\tau = \alpha(-1, c)$, and $\Gamma = \langle \sigma_1, \tau \rangle \leq \text{AGL}(1, \mathbb{Z}_q)$. Then

- (i) $\Gamma \cong (C_q \rtimes C_p) \rtimes C_2$ is the group of a chiral $\mathcal{P}$ of type $\{q, 2p\}$.
- (ii) $\mathcal{P} \in \mathcal{T}$, and is basal in $\mathcal{T}$.
- (iii) If $m = 3$, then $\Gamma = \text{AGL}(1, \mathbb{Z}_{p^2})$.

Conjecture (Chiral polyhedra from $\text{AFL}(1, \mathbb{F}_q)$)

In progress...



Classify some basal families for the following sets:

- $\mathcal{C}_{\text{sqf}} = \{\text{chiral } \mathcal{P} \mid |\Gamma(\mathcal{P})| \text{ is squarefree}\}.$
- $\mathcal{C}_{\text{meta}} = \{\text{chiral } \mathcal{P} \mid \Gamma(\mathcal{P}) \text{ is metacyclic}\}.$
- Others?

Question

Do all basal polyhedra in $\mathcal{C}_{\text{meta}}$ arise from an AGL-like construction?



Develop theory for basality in general!

- Classify the families $\mathcal{U}$ for which each $\mathcal{P} \in \mathcal{U}$ has a unique basal terminus in $\mathcal{U}$,

$$\text{e. g., } \mathcal{U} = \{\mathcal{P} \in \mathcal{C}(\mathbb{Z}^+, 1) \mid \mathcal{P} \text{ of type } \{2^s, 2^t\}\}.$$

Break into higher ranks.

- Let $\mathcal{T}_d$ denote the set of tight chiral d -polytopes. In [6], it was shown that $\mathcal{T}_d = \emptyset$ for ranks $d \geq 5$ since basal $\mathcal{P}$ in $\mathcal{T}_d$ do not exist for those ranks!

Question

Is there a class $\mathcal{C}_d$ of chiral d -polytopes for which there are no basal $\mathcal{P}$ in $\mathcal{C}_d$ for all $d \geq N$?

THANK YOU!
Questions?

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